

# SABELO DLAMINI

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## PROFILE

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Computer Science graduate from the University of Cape Town (UCT) with a passion for software development and cybersecurity. Eager to contribute technical skills and dedication to innovative projects. Proficient in various platforms, languages, and embedded systems, with a keen interest in exploring security protocols and threat mitigation. Experienced with cutting-edge development tools and procedures, and enthusiastic about applying cybersecurity principles to enhance software resilience. Able to effectively self-manage during independent projects, as well as collaborate as part of a productive team.

## EDUCATION

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**Gugulesizwe Secondary School**  
*National Senior Certificate(Matric)*

January 2013 - December 2017

**University of Cape Town**

*Bachelor of Science in Computer Engineering and Computer Sciences*

January 2018 - December 2023

## EXPERIENCE

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**UVU Africa Capaciti**

*Trainee - Ardagh Accelerator Programme*

March 2024 - Present

- Offered comprehensive training, providing real-world project experience and exposure to various technologies and domains.
- Focused on data analytics and cloud computing, gaining hands-on experience in these fields.
- Benefited from mentorship and guidance, enhancing both technical and soft skills.
- Participated in regular performance reviews with constructive feedback, promoting continuous learning and career growth.

## TECHNICAL SKILLS

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**Programming Languages**

Python, SQL, HTML, JavaScript, Java, C, C#, C++, Kotlin, Verilog, VHDL, Assembly.

**Software**

MS Office, GitHub, Power BI, Power Automate, Microsoft Azure.

**Operating Systems**

Linux , Windows.

**Embedded Systems**

Microcontroller programming, hardware-software interfaces and Developed and operated Raspberry Pi OW

**Languages**

IsiZulu, English, Xhosa.

**Behavioral**

Team Collaboration, Problem Solving, Adaptability, Analytical Thinking.

# PROJECTS

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## 2D Median Filter for Image Smoothing

Completed

*Java*

*UCT*

- Implement two parallel filters for smoothing RGB colour images using parallel
- algorithm to show original image, applying the image with a mean filter (middle) and a median filter (right).
- A mean filter sets each pixel in the image to the average of the surrounding pixels,
- whereas a median filter sets each pixel to the median of the surrounding pixels.
- Both methods use a Sliding square window of a specified width  $w$  ( $w$  is an odd number  $\geq 3$ )
- that defines the neighbouring pixels that are used to calculate the mean or the median.

## Building a 'video' from a large image

Completed

*C++*

*UCT*

- Using C++ To create a video, we extract pixels from a large image to produce a video that
- captures the movement of a much smaller window across large image, focusing on a specific
- trajectory. we position a rectangular window within this large image and extract all the
- pixels that overlap it to produce one frame. By shifting the window's position, we
- generate a sequence of image frames converted into a video using a python program.

## Serverless Data Processing with Azure Functions

Completed

*Python*

*Capaciti*

- This project involved creating a serverless data processing solution using Azure Functions and Power Automate. I developed an Azure
- Function to process JSON data files, extracting valuable information and storing it in Azure Table Storage.
- Power Automate was configured to trigger the Azure Function whenever a new JSON file was uploaded to Azure Blob Storage.
- The processed data was then summarized and sent via email.
- This project showcased my skills in serverless computing, data processing, and the integration of Azure Functions with Power Automate for efficient data workflows.

# INTERESTS

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Software Development

Networking

Cloud Computing

Machine Learning

Cyber-security

# REFERENCES

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